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Track Reconstruction using Masked Autoencoders and Graph Neural Networks

CMS End-to-End Deep Learning — Physics-Constrained Unsupervised Track Representation Learning

Organization: ML4SCI (Machine Learning for Science) | Contact: ml4-sci@cern.ch

Project Title	Track Reconstruction using Masked Autoencoders and Graph Neural Networks
Corresponding Project	E2E – End-to-End Deep Learning (proposal_E2E2 / proposal_E2E6)
Organization	ML4SCI – Machine Learning for Science (CERN affiliated)
Applicant Name	Ravakutam Mounika
Applicant Background	B.Tech – Robotics & Automation Engineering, 6th Semester, REVA University, Bangalore | CGPA: 9.22/10
Applicant GitHub	https://github.com/Mounika0021
Evaluation Tasks	Task 2a (MAE) and Task 2b (GNN) — both completed prior to proposal
Project Size	175-Hour Medium Project
Primary Languages	Python – PyTorch, PyTorch Geometric, uproot
Preferred Mentors	Diptarko Choudhury (NISER); Ruchi Chudasama (Univ. Alabama); Sergei Gleyzer (Univ. Alabama)

1. Abstract

State-of-the-art particle identification at the CERN CMS experiment has historically relied on hand- engineered high-level features — reconstructed jet masses, isolation variables, b-tagging discriminants
— derived from the raw detector readout through a lengthy reconstruction chain. While physically interpretable, this pipeline discards a substantial fraction of the low-level kinematic information present in the original track-level data and introduces biases that are difficult to characterise systematically.

This proposal describes a 12-week research and development effort to advance End-to-End (E2E) deep learning for the CMS experiment at the track-representation level. Rather than operating on pixelised calorimeter images, the proposed work trains models directly on variable-length sequences of particle 4- momenta (E, px, py, pz) — a representation that is both more physically natural and closer to the true information content of the CMS tracking detector. Two complementary architectures will be developed and evaluated: a Masked Autoencoder (MAE) applying self-supervised masking to sets of particle tracks, and an EdgeConv Graph Neural Network (GNN) encoding local particle interaction structure via message passing over a k-nearest-neighbour graph.

A central scientific contribution is the systematic comparison between set-based Transformer models and graph-based inductive models for particle reconstruction, motivated by preliminary evaluation tasks (Tasks 2a and 2b) already completed by the

applicant. Those experiments identified a concrete failure mode in the standard MAE objective — collapse to trivial near-zero predictions without physics constraints — and demonstrated graph-based inductive bias as a partial remedy. The proposed project builds on these empirical findings by integrating physics-informed loss functions (jet mass consistency, 4-momentum conservation), developing hybrid MAE+GNN architectures, and benchmarking all models using the invariant jet mass as the primary physics-grounded metric.

The preliminary MAE implementation (Task 2a) revealed a critical failure mode: collapse to near-zero predictions under standard reconstruction objectives. This behaviour highlights a fundamental limitation of dense architectures when applied to inherently sparse particle-level data, where meaningful signal is concentrated in a small subset of tracks.

This observation motivates the exploration of Sparse Neural Networks as a principled solution, where computation is restricted to physically relevant regions of the latent manifold. By introducing sparsity as an inductive bias, the proposed work aims to improve both reconstruction fidelity and computational efficiency, aligning with the requirements of CMS-scale data processing and potential deployment within latency-constrained environments such as the High-Level Trigger.

This work positions sparsity not as an optimisation trick, but as a fundamental inductive bias required for physically meaningful learning in high-energy physics data.

2. Scientific Background and Motivation

2.1 The CMS Detector and the E2E Paradigm

The Compact Muon Solenoid (CMS) detector at CERN's Large Hadron Collider records the products of proton-proton collisions at centre-of-mass energies up to 13.6 TeV. Charged particles traversing the inner silicon tracking detector leave sequential position measurements from which 3-momenta are reconstructed; combined with calorimeter energy deposits, each reconstructed track is associated with a 4-momentum vector (E, px, py, pz) — the natural Lorentz-covariant representation of a particle's kinematic state. The E2E philosophy posits that a sufficiently expressive neural network trained directly on low-level representations can outperform feature-engineered classifiers by accessing information discarded during conventional reconstruction.

2.2 Why Track-Level Representation Learning Is Non-Trivial

Track multiplicity varies event-by-event — from O(10) to O(100+) constituents per jet — requiring either fixed-length padding or a truly permutation-invariant architecture. The 4-momentum components span multiple orders of magnitude and are constrained by physical relationships (on-shell condition, energy positivity) that a naive MSE objective does not enforce. Without physics constraints, reconstruction models are susceptible to trivial solutions — predicting the distribution mean for all masked positions — satisfying the loss while carrying no physical content.

2.3 Electron/Photon and Quark/Gluon Classification Benchmarks

Two physics classification benchmarks validate the quality of learned track representations downstream. Electron vs. Photon discrimination exploits the presence or absence of a matched charged-particle track: electrons produce ECAL shower clusters co-localised with a tracker hit, photons produce identical calorimeter signatures with no corresponding track. Quark-initiated vs. Gluon-initiated jet classification probes multi-particle substructure: gluon jets are broader and of higher multiplicity due to the larger colour charge of the gluon, a difference that manifests at the level of individual track kinematics and angular distributions.

3. System Architecture and Pipeline

Figure 1 below illustrates the complete end-to-end pipeline from raw ROOT data to physics-grounded evaluation and downstream classification. Each stage is described in detail in section 4.

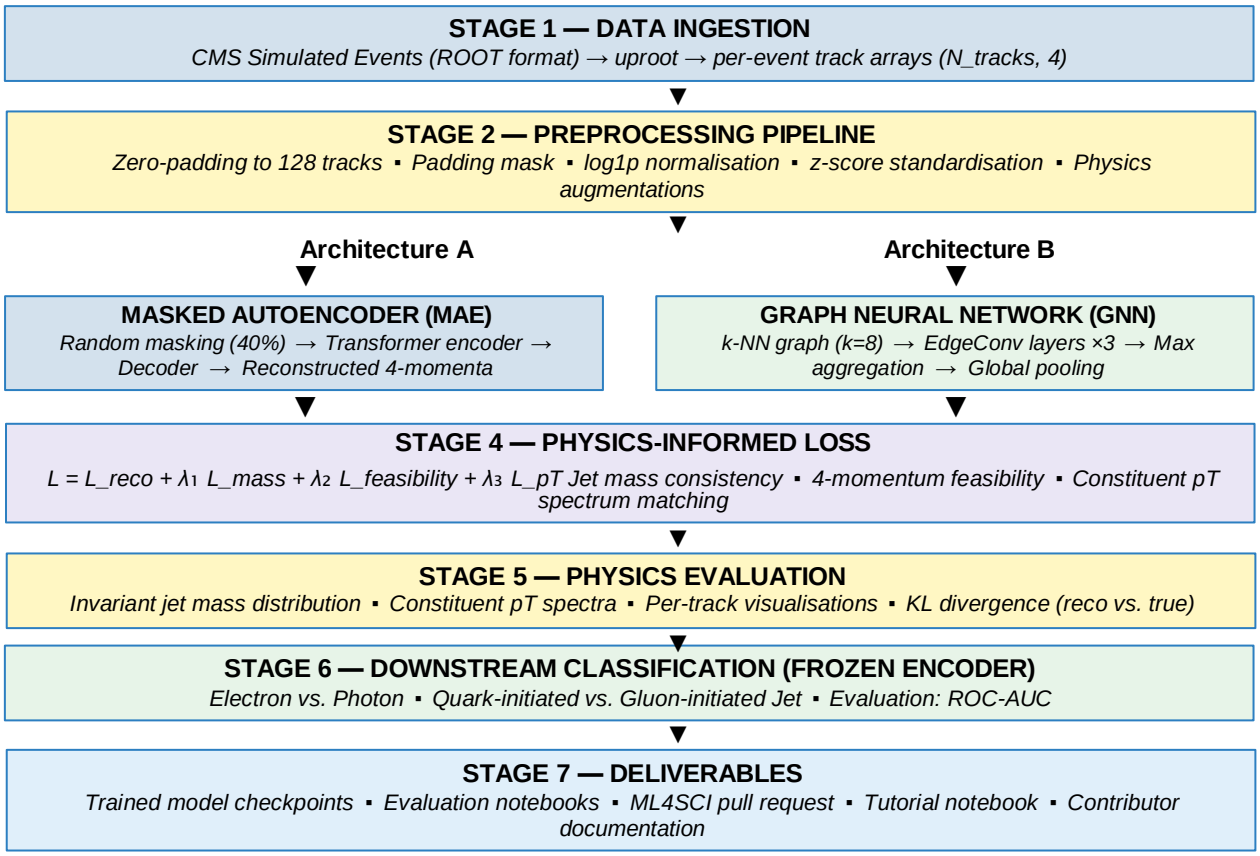

Figure 1. End-to-end pipeline from CMS ROOT data to physics evaluation and downstream benchmarks

3.1 Data Representation and Preprocessing

3.1.1 Track-Level Input Representation

Raw CMS simulated event data is stored in ROOT format and accessed via uproot. The preprocessing pipeline proceeds as follows: ROOT files are read per event, extracting track arrays of shape (N_tracks, 4); variable-length sets are zero-padded to 128 tracks per event yielding (B, 128, 4) tensors; a binary padding mask (B, 128) marks genuine tracks vs. padded slots and is propagated through all attention and pooling operations; log1p compression E_scaled = log(1 + E) is applied per-component to compress the multi-decade dynamic range; global z-score normalisation using training-set statistics is applied after log1p, excluding padded positions.

3.1.2 Augmentation Strategy

Random rotation in the transverse plane (px, py) by multiples of 90 degrees respects the approximate phi-symmetry of CMS. Azimuthal reflection (px, py) → (px, −py) is an exact symmetry of the collision process. No Gaussian noise is applied: particle track data has a physically motivated zero-signal state that artificial noise would violate.

4. Technical Implementation Plan

4.1 Architecture A — Masked Autoencoder (MAE)

Component	Design
Input	(B, 128, 4) normalised track tensor; padding mask propagated through all operations
Masking	40% of genuine tracks replaced with learned [MASK] embedding; masking rate is a tunable hyperparameter
Encoder	Transformer encoder with multi-head self-attention and gated feed- forward blocks; no positional encoding (permutation invariance by design); padding mask prevents attention to padded slots
Decoder	Lightweight Transformer decoder; reconstructs 4-momentum vectors of masked tracks from encoded visible-token context
Objective	MSE at masked positions + physics-informed auxiliary loss terms (Section 4.4); physics terms are the primary novel contribution

4.2 Architecture B — Graph Neural Network (GNN)

Component	Design
Graph construction	k-NN graph (k=8) in normalised (E, px, py, pz) space per event; padded tracks excluded from edge construction
Edge features	[x_i, x_j − x_i] concatenation: absolute node features and relative offset to each neighbour, capturing local geometric structure in momentum space
Message passing	3 EdgeConv layers with max aggregation over neighbourhood; residual connections between layers; batch normalisation after each layer
Pooling	Global mean pooling over genuine track nodes (padding mask applied) yielding a permutation-invariant jet-level embedding
Objective	Per-track 4-momentum reconstruction MSE + physics-informed auxiliary terms; identical loss framework as MAE to enable direct comparison

4.3 Physics-Informed Loss Functions (Core Technical Contribution)

The central technical contribution is the design and systematic evaluation of differentiable physics- informed loss terms that prevent trivial solutions and guide reconstruction toward physically consistent outputs:

- Jet mass consistency: $L_mass = (M_reco - M_true)^2$, where M_reco is computed from the sum of reconstructed track 4-momenta via $M^2 = (\Sigma E_i)^2 - |\Sigma p_i|^2$. Directly penalises the most consequential failure mode identified in preliminary experiments.
- 4-momentum feasibility: Smooth ReLU penalties on $E_i < 0$ (negative energy) and $|p_i|/E_i > 1$ (superluminal tracks); enforces basic kinematic consistency without hard constraints that would complicate gradient flow.
- Constituent pT spectrum matching: KL divergence between predicted and true distributions of constituent transverse momenta within a minibatch; discourages collapse to a single kinematic configuration.

Total objective: $L = L_reco + \lambda_1 L_mass + \lambda_2 L_feas + \lambda_3 L_pT$. Coefficients $\{\lambda_i\}$ are tuned via validation- set jet mass KL divergence. An ablation study will systematically evaluate each term's contribution to eliminating degeneracy.

4.4 Evaluation Metrics

Metric	Justification
Invariant jet mass	Physics-grounded scalar summary; computed from sum of
(primary)	predicted 4- momenta; compared as distribution via KL divergence and histogram overlay
Constituent pT spectrum	Per-track kinematic diagnostic; sensitive to systematic biases invisible to jet mass; reported as distribution over all test-set tracks
ROC-AUC (downstream)	Classifier performance on E/Photon and Q/G benchmarks using frozen learned embeddings; tests physical utility of representations
Reconstruction MSE	Per-component MSE at masked/predicted positions; used for optimisation monitoring and ablations, not as primary quality metric

FLOPS vs. Physics Accuracy (Pareto Frontier Analysis)

To align with CMS deployment constraints, models will be evaluated not only on reconstruction fidelity but also on computational efficiency. An explicit Pareto analysis will be conducted, comparing floating- point operations (FLOPS) against physics performance metrics such as jet mass KL divergence and downstream ROC-AUC. This enables identification of optimal operating points balancing accuracy and efficiency.

5. Empirical Motivation from Evaluation Tasks

Prior to this proposal the applicant completed ML4SCI evaluation tasks 2a (Track-level MAE) and 2b (EdgeConv GNN), providing the empirical foundation that motivates the research direction above. Key findings are summarised here as supporting evidence.

5.1 Task 2a — MAE: The Trivial Solution Problem

A track-level MAE was implemented with a Transformer encoder and trained on standard MSE over particle 4-momenta padded to 128 tracks per event (ROOT → uproot → PyTorch pipeline). Training was numerically stable. However, qualitative analysis revealed that the decoder converged to near-zero predictions for all masked tracks — a degenerate solution minimising MSE at the cost of all physical content. Jet mass distributions computed from reconstructed tracks were severely underestimated. This failure is directly attributable to the absence of physics-informed loss terms, and is precisely the problem Section 4.4 is designed to solve.

5.2 Task 2b — GNN: Inductive Bias as Partial Remedy

An EdgeConv GNN (k=8) trained on the same dataset showed faster convergence and produced non- trivial per-track predictions — no collapse to zero outputs. Jet mass distributions showed improved alignment with the true distribution. This is attributed to the EdgeConv inductive bias: encoding relative 4-momentum differences between neighbouring particles provides a prior structurally aligned with collinear radiation patterns of jet fragmentation. Despite this, reconstruction fidelity in distribution tails remains limited without explicit physics constraints.

5.3 Key Comparative Insight

Graph-based inductive biases provide a meaningful advantage over permutation-invariant set-based models for particle track reconstruction, but neither architecture achieves physically faithful reconstruction without explicit physics constraints in the training objective. This empirically grounded conclusion directly shapes the proposed research

agenda: physics-informed loss integration as the primary technical contribution, hybrid MAE+GNN architectures as the primary architectural innovation.

5.4 From Dense Reconstruction to Sparse Representations

The empirical findings from Tasks 2a and 2b suggest that the primary limitation of current approaches is not model capacity, but the mismatch between dense computation and sparse physical structure.

In CMS track data, meaningful signal is inherently sparse: only a subset of particle tracks contributes significantly to jet-level observables, while the remainder forms a low-information background. Dense architectures such as Transformers process all inputs uniformly, allowing trivial solutions such as mean prediction collapse to dominate the optimisation landscape.

This motivates a shift toward Sparse Neural Networks, where computation is explicitly restricted to active regions of the input space. By leveraging submanifold sparse convolutions (e.g., MinkowskiEngine, SpConv), the model can operate directly on the sparse support of particle interactions, preserving high- resolution structure while significantly reducing computational overhead.

This transition represents a natural progression from the observed limitations of MAE and the partial success of GNNs, toward architectures that explicitly encode sparsity as a first-class inductive bias.

6. 12-Week Project Roadmap (175 Hours)

~14–15 hours per week. Evaluation tasks 2a and 2b are complete; this timeline covers the proposed project from Community Bonding onward.

Wk	Hrs	Activity	Deliverable
CB	10	Community Bonding: Discuss Task 2a/2b findings with mentors (D. Choudhury, R. Chudasama, S. Gleyzer); agree on physics loss priority and evaluation protocol; set up ML4SCI codebase and W&B experiment tracking	Scope alignment note; W&B project initialised
1	15	Data pipeline hardening: Extend ROOT/uproot loader to multiple jet samples and pileup conditions; validate padding mask propagation through attention and pooling; constituent multiplicity statistics notebook	Robust DataLoader; data statistics notebook
2	15	Jet mass loss: Implement differentiable M^2 computation from batched 4-momenta; integrate L_mass into MAE objective; ablation: MSE-only vs. MSE + L_mass; evaluate on validation jet mass KL divergence	Mass loss module; ablation results
3	14	Feasibility + pT constraints: Implement energy positivity and mass-shell penalties; constituent pT KL divergence term; joint tuning of loss coefficients; retrain MAE on full physics-informed objective	Physics-constrained MAE checkpoint
4	15	MAE architecture ablations: Huber loss vs. MSE per component; masking ratio sweep (30/40/50%); encoder depth ablation (2/4/6 layers); all evaluated on jet mass KL divergence	Architecture ablation report

5	15	GNN + physics loss: Apply identical physics-informed loss to GNN; compare MAE vs. GNN under same loss; isolate contribution of architecture vs. loss to reconstruction quality	GNN physics-loss checkpoint; comparison table
6	14	Hybrid MAE+GNN: Implement GNN-enriched track embedding as drop-in for linear projection in MAE encoder; initial training and loss comparison against standalone models	Hybrid model prototype; training curves
7	15	Hybrid evaluation: Full evaluation of hybrid vs. standalone MAE and GNN on jet mass, constituent pT, and per-event visualisations; identify remaining failure modes	Comparative evaluation notebook
8	14	Downstream classification: Freeze encoder weights; linear probe on E/Photon and Q/G tasks; report ROC- AUC to validate physical utility of learned representations	Downstream AUC results; benchmark table
9	15	Hyperparameter optimisation: Optuna search over {lr, loss coefficients, k, mask ratio, GNN depth, encoder dim}; 5-fold cross-validation; select final model configurations	Optimised checkpoints; CV results
10	14	Anomaly detection prototype: Reconstruction-error- based anomaly score; calibration on SM events; preliminary ROC on simulated BSM signal events	Anomaly detection notebook
11	15	Code integration: Refactor to ML4SCI module conventions; YAML config system; pytest unit tests for data pipeline, loss functions, model interfaces	Refactored codebase; passing test suite
12	14	Final submission: Tutorial Jupyter notebook (ROOT → training → evaluation); complete API docstrings; GSoC final report; submit PR to ML4SCI repository	Final PR; tutorial notebook; project report

7. Why I Am the Right Candidate

7.1 I Have Already Built and Debugged These Exact Models

The two architectures at the centre of this proposal have already been implemented, trained, and evaluated. This is not theoretical familiarity: the applicant encountered and resolved concrete engineering challenges — NaN losses in early MAE runs (traced to unnormalised log1p outputs in padded slots), silent shape mismatches in the padding mask during attention computation, and gradient flow failures through EdgeConv aggregation when processing variable-density graphs. Each was diagnosed by inspecting intermediate activations and fixed by targeted code changes.

More critically, the applicant understands not just that the MAE collapsed to near-zero predictions, but precisely why: the MSE objective over continuous, zero-centred normalised data has a degenerate minimum at the dataset mean, which gradient descent finds efficiently. This causal understanding — not just symptom observation — is what motivates the specific physics-informed loss design in Section 4.4. A candidate who has only read about MAEs would propose adding a physics loss; a candidate who has debugged one knows which term to add first and why.

7.2 Medical Imaging CNN: Transferable Methodology

The applicant's Pneumonia Detection pipeline from chest X-rays is structurally analogous to the track reconstruction problem: in both settings, the model must localise sparse,

high-dynamic-range signalsagainst a near-zero background where meaningful information is concentrated in a small fraction of the input space. Techniques applied there — log-space intensity normalisation, background-aware loss weighting, validation via domain-interpretable metrics — map directly onto log1p normalisation, physics- informed loss design, and jet-mass-based evaluation in the current proposal.

7.3 Robotics: Sensor Fusion and Resource-Constrained Engineering

Experience fusing heterogeneous sensor streams on the Snake Robot (Dynamixel servo torque, IMU orientation, contact switches) and the Bluetooth River Cleaning Robot cultivated a specific skill directly relevant here: reasoning about signal calibration, noise models, and failure modes in multimodal systems. Padded zero-tracks are silent sensors that must be masked without corrupting attention weights; log1p normalisation must be consistent across all four 4-momentum components without channel-specific biases. The habit of thinking about edge cases — not just the nominal path — is a direct product of robotics systems experience.

7.4 Framework and Infrastructure

All completed and proposed work uses PyTorch and PyTorch Geometric, consistent with the ML4SCI E2E codebase. The ROOT pipeline uses uproot and awkward-array. Experiment tracking uses Weights & Biases; all runs are logged with hyperparameters, loss curves, and output distribution plots. The applicant is familiar with GitHub Actions for CI and will maintain a clean, reviewable commit history throughout.

7.5 Communication Commitment

All mentor communication will be conducted exclusively through ml4-sci@cern.ch, in full compliance with ML4SCI community guidelines. The applicant commits to weekly written progress updates, immediate escalation of blockers, and transparent reporting of negative results — as demonstrated by the honest failure analysis in Section 5.

7.6 Resource-Constrained Perspective

All experiments have been developed and tested on a personal system (Intel i7 CPU with integrated graphics), without access to dedicated GPU clusters. This constraint motivated careful attention to computational efficiency and memory usage throughout model design. The exploration of Sparse Neural Networks is therefore not only scientifically motivated but also practically aligned with enabling high- energy physics models to run on commodity hardware and edge environments.

8. Preferred Mentors

The following mentors are listed on the official ML4SCI GSoC 2026 project pages for proposal_E2E2 (End-to-End event classification with sparse neural networks) and proposal_E2E6 (End-to-End particle collision track reconstruction). These are the two project pages most closely aligned with the work described in this proposal.

Mentor	Institution	Relevant Project
Diptarko Choudhury	NISER (National Institute of Science Education and Research)	E2E2 (primary) + E2E6

Ruchi Chudasama	University of Alabama	E2E2 + E2E6 (primary)
Sergei Gleyzer	University of Alabama	E2E2 + E2E6 (org admin)
Shravan Chaudhari	New York University	E2E6
Emanuele Usai	University of Alabama	E2E2
Eric Reinhardt	University of Alabama	E2E2

Important: Do not contact any mentor directly by email. All questions and the final proposal submission must go through the official address **ml4-sci@cern.ch** and the Google Form linked on the project pages.

9. Relationship to Existing Work

The E2E classification concept in HEP was introduced in CMS internal studies and formalised by Collaboration et al. (2017). Subsequent work has operated on pixelised calorimeter images with dense CNNs, or on reconstructed constituent lists with point-cloud models (PointNet++, DGCNN) and Transformers (Particle Transformer, ParT). The present proposal occupies a distinct position: it operates on raw track 4-momenta, focuses on unsupervised reconstruction rather than supervised classification, and introduces physics-informed loss terms as the primary contribution. The key novelties are: (1) a controlled empirical comparison of MAE and GNN paradigms on the same task under a common physics metric, (2) differentiable physics loss terms targeting empirically identified failure modes, and (3) hybrid MAE+GNN architectures. None of these contributions have been previously reported in the ML4SCI evaluation literature.

10. Future Directions Beyond Project Scope

- Sparse Neural Networks for CMS data: extending the current framework to submanifold sparse convolutional architectures (MinkowskiEngine / SpConv), enabling efficient processing of high- resolution detector data while maintaining physics fidelity. This directly addresses the dense- computation limitations observed in MAE training.
- Normalising flow decoder: replacing the MSE-trained decoder with a conditional normalising flow to model the multi-modal distribution of track 4-momenta and enable proper uncertainty quantification
- Anomaly detection at scale: deploying reconstruction-error-based anomaly scores on large- scale CMS Run 3 datasets, requiring careful calibration across pileup conditions
- CMS trigger integration: evaluating whether physics-constrained learned representations can be exported to TorchScript and deployed within the CMS High Level Trigger latency budget
- Extension to secondary vertices: applying the same graph-based framework to secondary vertex tracks, enabling end-to-end b-tagging without explicit vertex reconstruction

11. Selected References

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- ML4SCI E2E Repository: https://github.com/ML4SCI/E2EML